

Problem E. K-th one

Time Limit 1000 ms
Mem Limit 524288 kB

Description

In this challenge, you need to spice up a segment tree by adding a “find the k -th one” operation.

Input

The first line gives you two numbers n and m ($1 \leq n, m \leq 100000$), representing the size of the array and the number of operations. The next line lays out the initial state of the array with n and a_i ($a_i \in \{0, 1\}$). Then come the operations, each described as follows:

- 1 i : Flip the element at index i to its opposite.
- 2 k : Find the position of the k -th one (counting ones from zero, and rest assured, there are enough ones in the array).

Output

For every operation of the second type, print the index you found (remember, all indices in this problem start from 0).

Sample 1

Input	Output
5 7 1 1 0 1 0 2 0 2 1 2 2 1 2 2 3 1 0 2 0	0 1 3 3 1